AA256 HW3 Solution

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1 Problem 1

Use engineering shear strain with component order (x, y, xy). For each ply, the stress is related to mid-plane strain and curvature by:

$$\{\sigma\}_k = [Q]_k(\{\varepsilon^0\} + z\{\kappa\})$$

The local plane-stress stiffness is $[Q] = [S]^{-1}$, where:

$$[S] = \begin{bmatrix} \frac{1}{E_x} & \frac{-\nu_{xy}}{E_x} & 0 \\ \frac{-\nu_{xy}}{E_x} & \frac{1}{E_y} & 0 \\ 0 & 0 & \frac{1}{G_{xy}} \end{bmatrix}$$

$$[Q]_k = [T]_k{}^{-1}[Q][T]_k{}^{-T}$$

$$[T]_k = \begin{bmatrix} c^2 & s^2 & 2cs \\ s^2 & c^2 & -2cs \\ -cs & cs & c^2 - s^2 \end{bmatrix}$$

Here $c = \cos\theta_k$ and $s = \sin\theta_k$. The ply angles are measured counterclockwise from the global x direction. For n plies, $t = nh$ and $z_k = -t/2 + kh$, with the listed stack running from top to bottom and z positive downward.

$$[A] = \sum^n{}_{k=1}[Q]_k(z_k - z_{k-1})$$

$$[B] = \frac{1}{2}\sum^n{}_{k=1}[Q]_k(z_k{}^2 - z_{k-1}{}^2)$$

$$[D] = \frac{1}{3}\sum^n{}_{k=1}[Q]_k(z_k{}^3 - z_{k-1}{}^3)$$

$$\begin{bmatrix}\{N\} \\ \{M\}\end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix}\begin{bmatrix}\{\varepsilon^0\} \\ \{\kappa\}\end{bmatrix}$$

$$\begin{bmatrix} A & B \\ B & D \end{bmatrix}^{-1} = \begin{bmatrix} \alpha & \beta \\ \beta^T & \delta \end{bmatrix}$$

Invert the full 6 × 6 matrix, then extract its 3 × 3 blocks. In general $\alpha \neq A^{-1}$, $\beta \neq B^{-1}$, and $\delta \neq D^{-1}$. The lower-left block is β^T. For symmetric laminates, $B = \beta = 0$, $\alpha = A^{-1}$, and $\delta = D^{-1}$.

1.1 Layup A

[0/0/90/90]

Expanded stack (degrees): [0 / 0 / 90 / 90]

n = 4, h = 0.007 in, t = 0.028 in. Entries below are rounded; calculations use full precision.

$$[A] = 10^{6}\begin{bmatrix} 0.344513 & 0.010237 & 0 \\ 0.010237 & 0.344513 & 0 \\ 0 & 0 & 0.025200 \end{bmatrix}\ \text{lb/in}$$

$$[B] = 10^{3}\begin{bmatrix} -2.155670 & 0 & 0 \\ 0 & 2.155670 & 0 \\ 0 & 0 & 0 \end{bmatrix}\ \text{lb}$$

$$[D] = 10^{3}\begin{bmatrix} 0.022508 & 0.000669 & 0 \\ 0.000669 & 0.022508 & 0 \\ 0 & 0 & 0.001646 \end{bmatrix}\ \text{lb·in}$$

$$[\alpha] = 10^{-6}\begin{bmatrix} 7.259255 & -0.215704 & 0 \\ -0.215704 & 7.259255 & 0 \\ 0 & 0 & 39.682540 \end{bmatrix}\ \text{in/lb}$$

$$[\beta] = 10^{-3}\begin{bmatrix} 0.695238 & 0 & 0 \\ 0 & -0.695238 & 0 \\ 0 & 0 & 0 \end{bmatrix}\ \text{1/lb}$$

$$[\delta] = 10^{-3}\begin{bmatrix} 111.111047 & -3.301585 & 0 \\ -3.301585 & 111.111047 & 0 \\ 0 & 0 & 607.385811 \end{bmatrix}\ \text{1/(lb·in)}$$

This unsymmetric stack has extension–bending coupling. Reversing the z convention reverses the signs of B and β, while A, D, α, and δ are unchanged.

1.2 Layup B

[+30/−30/−30/+30]

Expanded stack (degrees): [30 / −30 / −30 / 30]

n = 4, h = 0.007 in, t = 0.028 in. Entries below are rounded; calculations use full precision.

$$[A] = 10^{6}\begin{bmatrix} 0.392036 & 0.116691 & 0 \\ 0.116691 & 0.084083 & 0 \\ 0 & 0 & 0.131654 \end{bmatrix} \text{ lb/in}$$

$$[B] = 10^{3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ lb}$$

$$[D] = 10^{3}\begin{bmatrix} 0.025613 & 0.007624 & 0.009546 \\ 0.007624 & 0.005493 & 0.003522 \\ 0.009546 & 0.003522 & 0.008601 \end{bmatrix} \text{ lb·in}$$

$$[\alpha] = 10^{-6}\begin{bmatrix} 4.346064 & -6.031457 & 0 \\ -6.031457 & 20.263400 & 0 \\ 0 & 0 & 7.595685 \end{bmatrix} \text{ in/lb}$$

$$[\beta] = 10^{-3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ 1/lb}$$

$$[\delta] = 10^{-3}\begin{bmatrix} 86.108912 & -78.960958 & -63.225718 \\ -78.960958 & 319.262763 & -43.115335 \\ -63.225718 & -43.115335 & 204.083621 \end{bmatrix} \text{ 1/(lb·in)}$$

Symmetry makes B and β exactly zero; any tiny computed values are floating-point roundoff.

1.3 Layup C

[45/−45]$_s$

Expanded stack (degrees): [45 / −45 / −45 / 45]

n = 4, h = 0.007 in, t = 0.028 in. Entries below are rounded; calculations use full precision.

$$[A] = 10^{6}\begin{bmatrix} 0.202575 & 0.152175 & 0 \\ 0.152175 & 0.202575 & 0 \\ 0 & 0 & 0.167138 \end{bmatrix}\ \text{lb/in}$$

$$[B] = 10^{3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\ \text{lb}$$

$$[D] = 10^{3}\begin{bmatrix} 0.013235 & 0.009942 & 0.007545 \\ 0.009942 & 0.013235 & 0.007545 \\ 0.007545 & 0.007545 & 0.010920 \end{bmatrix}\ \text{lb·in}$$

$$[\alpha] = 10^{-6}\begin{bmatrix} 11.330076 & -8.511193 & 0 \\ -8.511193 & 11.330076 & 0 \\ 0 & 0 & 5.983071 \end{bmatrix}\ \text{in/lb}$$

$$[\beta] = 10^{-3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\ \text{1/lb}$$

$$[\delta] = 10^{-3}\begin{bmatrix} 191.059140 & -112.633766 & -54.187143 \\ -112.633766 & 191.059140 & -54.187143 \\ -54.187143 & -54.187143 & 166.457625 \end{bmatrix}\ \text{1/(lb·in)}$$

Symmetry makes B and β exactly zero; any tiny computed values are floating-point roundoff.

1.4 Layup D

$[0/60]_s$

Expanded stack (degrees): [0 / 60 / 60 / 0]

n = 4, h = 0.007 in, t = 0.028 in. Entries below are rounded; calculations use full precision.

$$[A] = 10^{6}\begin{bmatrix} 0.368275 & 0.063464 & 0.035943 \\ 0.063464 & 0.214298 & 0.097404 \\ 0.035943 & 0.097404 & 0.078427 \end{bmatrix} \text{ lb/in}$$

$$[B] = 10^{3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ lb}$$

$$[D] = 10^{3}\begin{bmatrix} 0.037986 & 0.001538 & 0.000587 \\ 0.001538 & 0.005292 & 0.001591 \\ 0.000587 & 0.001591 & 0.002516 \end{bmatrix} \text{ lb·in}$$

$$[\alpha] = 10^{-6}\begin{bmatrix} 2.873518 & -0.579577 & -0.597120 \\ -0.579577 & 10.832186 & -13.187697 \\ -0.597120 & -13.187697 & 29.403205 \end{bmatrix} \text{ in/lb}$$

$$[\beta] = 10^{-3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ 1/lb}$$

$$[\delta] = 10^{-3}\begin{bmatrix} 26.644453 & -7.255063 & -1.629671 \\ -7.255063 & 235.315432 & -147.116960 \\ -1.629671 & -147.116960 & 490.907201 \end{bmatrix} \text{ 1/(lb·in)}$$

Symmetry makes B and β exactly zero; any tiny computed values are floating-point roundoff.

1.5 Layup E

[0/45/−45/90]

Expanded stack (degrees): [0 / 45 / −45 / 90]

n = 4, h = 0.007 in, t = 0.028 in. Entries below are rounded; calculations use full precision.

$$[A] = 10^{6}\begin{bmatrix} 0.273544 & 0.081206 & 0 \\ 0.081206 & 0.273544 & 0 \\ 0 & 0 & 0.096169 \end{bmatrix}\ \text{lb/in}$$

$$[B] = 10^{3}\begin{bmatrix} -1.616753 & 0 & -0.269459 \\ 0 & 1.616753 & -0.269459 \\ -0.269459 & -0.269459 & 0 \end{bmatrix}\ \text{lb}$$

$$[D] = 10^{3}\begin{bmatrix} 0.021349 & 0.001828 & 0 \\ 0.001828 & 0.021349 & 0 \\ 0 & 0 & 0.002806 \end{bmatrix}\ \text{lb·in}$$

$$[\alpha] = 10^{-6}\begin{bmatrix} 9.930590 & -4.016564 & 2.915993 \\ -4.016564 & 9.930590 & -2.915993 \\ 2.915993 & -2.915993 & 12.342343 \end{bmatrix}\ \text{in/lb}$$

$$[\beta] = 10^{-3}\begin{bmatrix} 0.765258 & 0.275453 & 0.568010 \\ -0.275453 & -0.765258 & 0.568010 \\ 0.346903 & 0.346903 & 0 \end{bmatrix}\ \text{1/lb}$$

$$[\delta] = 10^{-3}\begin{bmatrix} 107.800855 & 16.008091 & 47.043053 \\ 16.008091 & 107.800855 & -47.043053 \\ 47.043053 & -47.043053 & 465.543127 \end{bmatrix}\ \text{1/(lb·in)}$$

This unsymmetric stack has extension–bending coupling. Reversing the z convention reverses the signs of B and β, while A, D, α, and δ are unchanged.

2 Problem 2

Take axis 1 along the tensile load, at +15° to the original 0° reference direction. Express every ply angle relative to this axis:

$$\theta'_k = \theta_k - 15°$$

Thus layup A becomes [−15/−15/75/75]. Apply the same subtraction to every ply of B, C, D, and E, then recompute the laminate stiffness and its full inverse in the load frame.

For a sufficiently long specimen under uniaxial force, with no applied moment and bending free to occur:

$$\{N'\} = \begin{bmatrix} N_1 \\ 0 \\ 0 \end{bmatrix}, \quad \{M'\} = \{0\}$$

$$\{\varepsilon^{0\prime}\} = [\alpha']\{N'\}, \quad \{\kappa'\} = [\beta']^{\mathrm{T}}\{N'\}$$

$$\varepsilon_1{}^0 = \alpha_{11}' N_1, \quad \sigma_1 = \frac{N_1}{t}$$

$$E_1 = \frac{\sigma_1}{\varepsilon_1{}^0} = \frac{1}{t\alpha_{11}'}$$

Layup	t (in)	α_{11}' (in/lb)	E_1 (msi)
A	0.028	8.80504395e-06	4.056117
B	0.028	3.59503045e-06	9.934349
C	0.028	9.22385957e-06	3.871946
D	0.028	3.59503045e-06	9.934349
E	0.028	1.02212547e-05	3.494120

For layups A and E, using A'^{-1} instead of the upper-left block of the full inverse would suppress extension–bending coupling and give incorrect moduli under the stated free-bending assumption. For the symmetric layups B, C, and D, $\alpha' = A'^{-1}$.

Checks: A, B, and D are symmetric; the assembled laminate stiffness times its inverse is the identity; B and β vanish for B–D. The tensile moduli also agree with transforming α directly into the load frame.